MANKIRAT SINGH

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PROFILE

Data graduate (MSc Bioinformatics and Computational Biology, University College Cork; thesis defended October 2026, results November 2026, graduation March 2027), working across biology and data. My largest projects are about the Irish electricity grid: a 2040 capacity and investment model for Cork and Dublin (demand projected for 713 electoral divisions, 5,000-run Monte Carlo cost ranges of €194–346m and €379–817m), a Power BI readiness dashboard for 178 substations with 27 DAX measures, and a day-ahead demand forecast on EirGrid's 15-minute system data that is 2.74% off per hour and 34% better than the industry benchmark. I build automated, tested pipelines, state my assumptions and uncertainty, and had my key clinical-trial results independently re-checked. I am looking for graduate roles in data, analytics and bioinformatics.

EDUCATION

MSc, Bioinformatics and Computational Biology (NFQ Level 9) **Sep 2025 – Present**
University College Cork, Ireland
Status: all modules completed and thesis defended (October 2026); final results November 2026
Modules: Data Mining, Introduction to Probability and Statistics, Data Analysis I & II, Introduction to Relational Databases (SQL), Programming for Bioscientists I & II (Python), Open Source Infrastructure for Modelling with Data (computer science, information systems and mathematics content)
Thesis: multi-omics exploratory analysis of gut microbiota and metabolome changes after bariatric surgery, IBIMA Plataforma BIONAND, Málaga

PG Diploma, Bioinformatics **Jul 2024 – Aug 2025**
St. Xavier's College, Mumbai, India
Grade: CGPA 8.50 / 10
Final project: Glypican-3 as a liver-cancer biomarker (TCGA, GEPIA2, protein structure)

BSc, Life Sciences **Jul 2020 – Jul 2023**
University of Delhi, New Delhi, India
Grade: CGPA 6.97 / 10, First Division

DATA & ANALYTICS PROJECTS

All-Island Wind Turn-Down from Public EirGrid Data (report · code) **Python, statistics**
- Estimated how much wind the all-island grid turns down when demand is low, from four years of public 15-minute EirGrid data (2022–2025): among hours with the same forecast wind, the quietest quarter by demand produced up to 23% less wind; the estimate tracks EirGrid's published curtailment (r = 0.86, mean difference 0.9 points)
- Checked it with a placebo on low-wind hours (about zero) and 95% intervals from resampling days, and replaced my first method after it failed its own placebo test; hours at the SNSP limit rose from 1.8% to 4.4% and peak at midday, while 61% of the turned-down wind is lost at night

Grid Readiness Dashboard to 2040 (dashboard · code) **Power BI, DAX, Power Query**
- Built a three-page Power BI readiness dashboard for 178 Cork and Dublin electricity substations on a star-schema model (6 tables, 11,000 forecast rows) with 27 DAX measures: KPI cards, trend lines, driver breakdown and slicers
- **Readiness progression:** tracks how many stations run out of spare capacity each year under ESB Networks' four forecast scenarios and compares it with my model's 2040 result (Dublin: 77 stations short vs 70 in ESB Networks' base case)
- **Performance scorecard:** red/amber/green site table (action needed, shortfall in MW, year the station is full) with conditional formatting, plus a bubble map of where the shortfalls are
- **Data validation:** found that ESB Networks' tables report 0 MW at every station for 2036–2038 in two scenarios, a gap rather than real data, and excluded it instead of counting 113 stations as full; the report and data model rebuild from code

Cork & Dublin 2040 Grid Investment Plans (Cork report · Dublin report · code) **Python, Monte Carlo**
- Answered a planning question for an infrastructure owner: how much power Cork and Dublin will need by 2040, which substations run out of room and what the new capacity would cost, from CSO, Census 2022 and ESB Networks data
- Cork: peak +91%, 45 of 65 stations short, €194–346m; Dublin: peak doubles, 77 of 113 short, €379–817m (5,000-run Monte Carlo). Recommended flexible demand as the cheaper first step, saving ~€68m / ~€203m, and named data-centre growth as the key uncertainty

Clinical-Trial Risk Screening Tool (report · code) **Python, scikit-learn**
- Analysed 40,124 Phase 2/3 trials from the ClinicalTrials.gov API to show sponsors why 1 in 6 stop early: mostly low enrolment, not the science
- Built a design-stage risk score tested on later, unseen years (AUC 0.67 vs 0.57 for a rule of thumb; its riskiest 10% stopped twice as often) and an interactive calculator that explains each trial's score; removed data leakage that would have inflated results

Clinical Trial Safety Analysis with Independent QC (report · code) **R, SQL (DuckDB)**
- Took a 254-patient medical-product trial from raw CDISC data to analysis datasets, safety tables and survival analysis, following a written specification for every derived variable
- Re-built the key results in a second, independent program: matched production on all 306 subjects and 354 adverse-event cells, with unit tests and structural checks

Irish Electricity Demand Forecasting (report · code) **Python, scikit-learn**

- Automated pipeline for 15-minute EirGrid data (2019–2026) with retries, caching and quality checks for gaps and stuck values; day-ahead forecast 2.74% off per hour on unseen data, 34% better than the industry benchmark

Health and Deprivation Across Ireland (report · code) **Python, regression**

- Census 2022 analysis of 3,420 communities via the CSO API with 13 automated quality checks and county-held-out validation: social factors explained 63% of the variation in poor health, age alone 1%

RELEVANT EXPERIENCE

Intern-Researcher, R&D (diagnostics product development) **Sep 2024 – Jan 2025**

Premas Life Sciences Pvt Ltd | New Delhi, India

- Developed NGS diagnostic kits for *Mycobacterium tuberculosis* and 3 other infectious diseases
- Improved the Illumina sequencing workflow, making it 20% more efficient, and built bioinformatics pipelines to validate each kit's sensitivity

Master Thesis Researcher **May 2026 – Oct 2026**

IBIMA Plataforma BIONAND | Málaga, Spain

- Built a reproducible multi-omics pipeline for a longitudinal bariatric-surgery cohort: 16 patients (8 sleeve gastrectomy, 8 gastric bypass) before and 3 months after surgery, plus 8 healthy lean controls; raw Ion Torrent 16S reads processed in QIIME2 (DADA2, SILVA) and LC-MS/MS faecal metabolomics analysed in R (phyloseq, ANCOM-BC2, mixOmics sPLS and DIABLO)
- Showed why study design matters: the surgery effect on the microbiome was significant with within-patient (paired) change ($p = 0.036$) but hidden in an unpaired test ($p = 0.378$); a group of 26 taxa tracked larger drops in fasting glucose ($\rho = -0.83$), and I reported that no procedure-specific signature survived cross-validation

Intern-Researcher **Jul 2024 – Nov 2024**

National Agri-Food Biotechnology Institute (NABI) | Mohali, India

- Analysed *Cyp19a1* expression across human and mouse models; ran in vivo studies on 40+ subjects and quantified behaviour from ANY-maze video-tracking data

Intern-Researcher **Jul 2023 – Nov 2023**

National Agri-Food Biotechnology Institute (NABI) | Mohali, India

- Studied the banana DNA-damage gene *Ma-SOG1* and designed expression vectors for *Agrobacterium*-mediated transformation; later followed the gene across 17 plant genomes in an automated comparative-genomics pipeline

TECHNICAL SKILLS

- **Data science & analytics:** Python (pandas, NumPy, scikit-learn), R, SQL (DuckDB), Microsoft Excel workbook parsing, Git & GitHub; analytical problem-solving from business question to data solution
- **Reporting & management information:** Microsoft Power BI (DAX, Power Query, star-schema data modelling, conditional formatting), KPI definition, business performance scorecards, interactive digital reports and maps
- **Data governance, quality & compliance:** automated quality checks, data fingerprints and logs, documented assumptions, independent reviews (double programming), CDISC regulatory data standards, reproducible pipelines
- **Quantitative analysis & risk:** Monte Carlo cost and risk ranges, investment cost modelling, risk assessment and classification models (gradient boosting, random forests, logistic regression), survival analysis, regression, forecasting
- **Responsible AI:** leakage checks, held-out and later-year testing, uncertainty and prediction intervals, clearly stated limitations
- **Communication & deliverables:** plain-language reports, recommendations and presentation slides for non-technical stakeholders; written specifications

CERTIFICATIONS, COMMUNITY & INTERESTS

- **Certifications:** Kaggle – Intro to Machine Learning, Data Visualization, Python, Pandas, Intro to Programming; Decode Life – AI and Machine Learning for Drug Discovery
- **Community & sports:** charity volunteer with EHSAAS Foundation; hosted the “STROKES” art society; GRSS roller-skating champion; former NCC cadet; photography featured in Photo Vogue
- **Languages: English (fluent, medium of instruction throughout), Hindi and Punjabi (native)**

REFEREES

- **Names of referees will be supplied on request.**